Praise SW

一、Lit Review

A. 心理学 / 理论基础

1. **Boothby & Bohns (2021), *Underestimating the Positive Impact of Our Compliments on Others***
   最关键的一篇。研究发现，人们会低估 compliments 对他人的积极影响，也会高估对方可能感到 awkward 的程度；而且在研究里，compliment givers 在表达后自己也感觉更好。这个几乎就是你项目的理论起点。 (Cornell eCommons)

2. **Ryan & Deci (2000), *Self-Determination Theory***
   SDT 认为人的基本心理需求包括 **autonomy, competence, relatedness**，这些需求被满足时，幸福感和内在动机会更高。你的装置如果要成立，不能只是“逼人夸人”，而要让用户感觉：这是我自愿说的、我说得出来、而且我和他人产生了连接。 (PubMed)

3. **Fredrickson (2001), *The Broaden-and-Build Theory of Positive Emotions***
   这篇经典理论认为正向情绪不只是“让人开心一下”，而是会扩展人的思维与行动倾向，并逐步积累 social / psychological resources。你可以用它说明：一个小小的真实赞美，可能不是一次性事件，而是 social openness 的起点。 (PEP实验室)

4. **Grant & Gino (2010), *A Little Thanks Goes a Long Way***
   虽然研究对象是 gratitude，不是 compliment，但很有用：被感谢的人会因为感到 **social worth** 和 **self-efficacy** 被增强，而更愿意继续 prosocial behavior。对你来说，这说明“积极表达”如果被回应或被看见，会形成正向循环。 (PubMed)

5. **JAMA Network Open (2023), *Prosocial Interventions and Health Outcomes: A Systematic Review and Meta-Analysis***
   系统综述发现，prosocial interventions 和更好的健康结果有关，而且这些干预可能同时让 giver 和 recipient 受益。你拿这篇可以证明：这不是空泛口号，prosocial behavior 确实可以被设计成 intervention。 (JAMA网络)

B. 数字干预 / HCI / 可设计性

6) *The Big Joy Project* / brief digital well-being intervention
Big Joy 是一个 7 天、每天 5–10 分钟的 web-based intervention，研究报告说它能提升 well-being、降低 stress、改善部分 health-related outcomes。它很适合你用来说明：**微小、低门槛、短时的积极行为练习** 是可以被设计、被规模化、也能产生 measurable impact 的。 (PMC)

6. **Ciocarlan et al. (2018), *Kindness is Contagious***
   这类 HCI 研究关注的是：怎样用 gamified / persuasive intervention 鼓励 random acts of kindness，改善 well-being。对你有两个启发：
   一是 prosociality 可以被“触发”；
   二是设计重点不只是内容，还包括 engagement mechanism。 (ACM Digital Library)
7. **“Orchestrating acts of kindness” (2025)**
   这篇做 diary study，整理出 everyday acts of kindness 对 actor 和 recipient 的影响因素，并把它们组织成面向设计的 framework。它特别适合你在 meeting 里说：我不只是在做一个 poetic object，我也在关注 kindness / praise 发生时的 context、characteristics、outcomes。 (科学直通车)

C. 形式 precedent / installation references

8. **Christine Wong Yap, *Complementary Compliments***
   这是一个关于 **drawing, coloring, and human connection** 的 installation，核心不是炫技，而是通过一个被设计的情境，让两个陌生人或同伴更容易彼此观察、表达善意、建立连接。它能证明：关于 praise / appreciation 的主题，本身就非常适合做 participatory installation。 (christinewongyap.com)
9. **WNDR Museum / Ziggy Yang, *Compliment Machine***
   这个 precedent 很直接：AI 生成 compliment，并把它变成一张可带走的 receipt。它对你最有价值的地方，不是“AI 会夸人”，而是它把 **无形语言 → 可见、可保存的 artifact**。这和你想把 spoken praise 转成视觉或声音反馈，非常接近。 (wndrmuseum.com)
10. **Rafael Lozano-Hemmer, *Voice Array* / *Open Air***
    如果你想讲 installation language，这个必须提。`Voice Array` 把参与者声音转成灯光脉冲并累积保存；`Open Air` 则让公众提交语音，让 24 盏 searchlights 根据声音的 frequency 和 amplitude 在天空中反应。它们说明：**voice 可以被空间化、可视化、公共化**，而且作品会因为参与者而持续生成。 (bitforms gallery)

二、Theoretical Framework

1.1 Problem layer：Why people do not actively express praise (Questionary)

A. Expression feels difficult

- awkward or unnatural 觉得自己说出来会尴尬、冒犯、太突兀。 (Cornell eCommons)
- shy / not good at expressing positive emotions
- do not know how to phrase praise

B. Social risk feels high

- fear of sounding fake
- fear of being seen as flattering
- concern about modesty / social norms

C. Appreciation does not automatically become language

- People may notice something positive but do not articulate it
- Many have not formed a habit of observing and voicing appreciation

D. Environment does not support deeper praise

- likes replace words
- Online interaction favors low-cost acknowledgement over sincere expression

E. Praise culture itself is fragile

- Some people also feel awkward receiving praise
- Everyday scripts for giving and receiving praise are underdeveloped

1.2 CORE* Mechanism layer：为什么一旦表达，会产生正向循环

1) Praise as connection

Giving sincere praise is not just saying something nice; it can function as an act of social connection. From the perspective of Self-Determination Theory, this may support the need for **relatedness**—**the, the** feeling of being connected to others and emotionally linked to them. When a person actively expresses appreciation, they may also feel more connected to themselves.

2) Praise as self-endorsed action

SDT also suggests that behaviors tend to lead to better psychological outcomes when they are experienced as voluntary rather than forced. This means the value of praise may be stronger when it feels like something the speaker genuinely wants to say, rather than something they are pressured to perform. This relates to **autonomy**.

3) Praise as a small success in expression

For many people, openly complimenting others is not easy. If a person successfully expresses sincere praise, that act may also produce a small sense of effectiveness or capability—related to **competence**. In this sense, the return to the giver may include not only warmth, but also a subtle feeling of agency. (Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being)

4) Praise as prosocial behavior

Sincere praise can also be understood as a form of **prosocial behavior**. Research suggests that prosocial behavior is positively associated with well-being, and that this relationship may be

mediated by the satisfaction of basic psychological needs. This helps explain why the benefit of praise may return to the giver rather than flow only outward.（Assessing the Relationship between Prosocial Behavior and Well-Being: Basic Psychological Need as the Mediator）

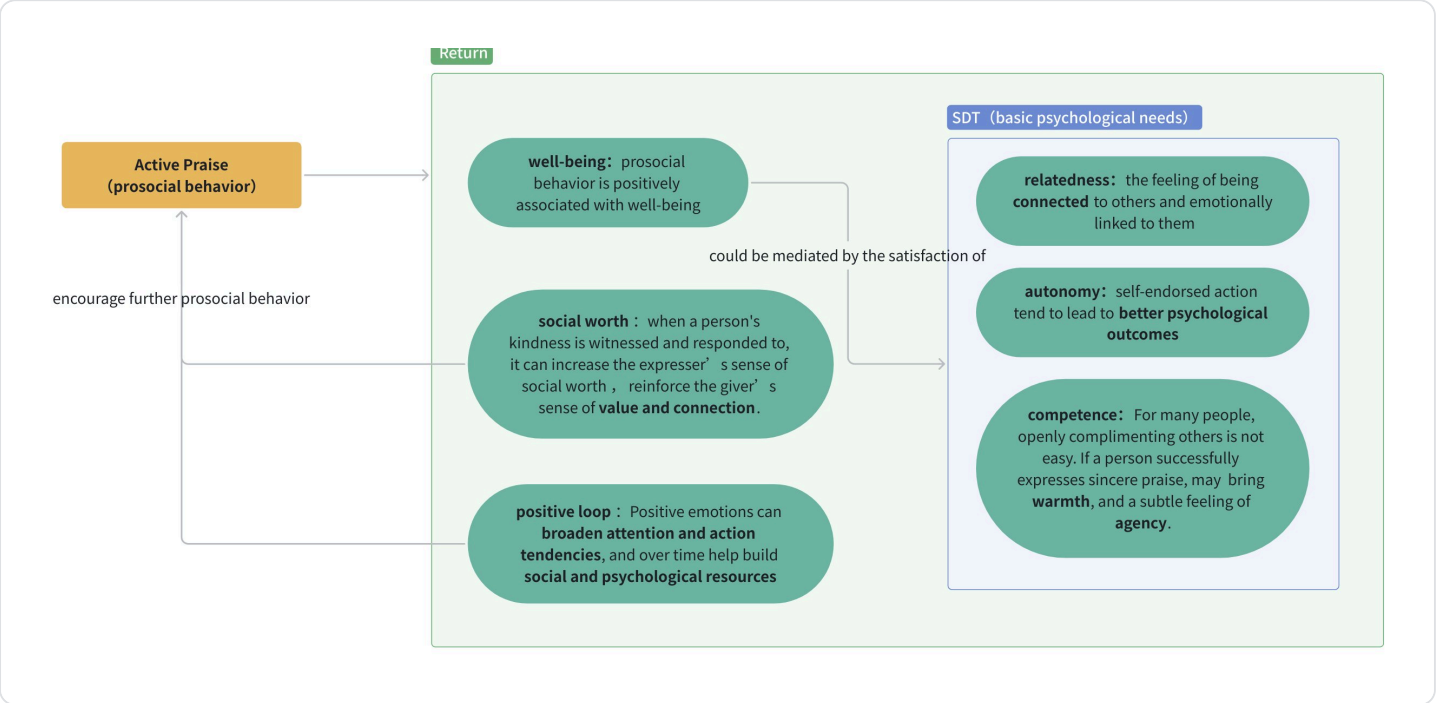
5) Praise as a source of social worth

Research on gratitude expression suggests that when a person's kindness is witnessed and responded to, it can increase the expresser’ s sense of **social worth** and encourage further prosocial behavior. Although gratitude is not identical to praise, this finding helps support the idea that expressing something positive to another person may also reinforce the giver’ s sense of value and connection. (A little thanks goes a long way: Explaining why gratitude expressions motivate prosocial behavior)

6) Positive emotion as an amplifier

The broaden-and-build theory does not specifically prove that praise itself causes positive emotion, but it helps explain what may happen if positive emotion is evoked through the act of praise. Positive emotions can broaden attention and action tendencies, and over time help build social and psychological resources. In this way, a small act of praise may become the beginning of a broader positive loop. (The Role of Positive Emotions in Positive Psychology)

Summary/ Basic Design Concept：Giving praise can benefit the giver, because expressing sincere positive regard can reduce social distance, satisfy relational needs, generate positive emotion, and reinforce one’ s sense of social value.



1.3 Design layer（features）

(a) Lower the threshold

降低开口成本，让人更容易说出一句真实赞美。这个回应的是 awkwardness barrier。 (Cornell eCommons)

SDT 认为人的基本心理需求包括 **autonomy, competence, relatedness**，这些需求被满足时，幸福感和内在动机会更高。你的装置如果要成立，不能只是“逼人夸人”，而要让用户感觉：这是我自愿说的、我说得出来、而且我和他人产生了连接。 (PubMed)

(b) Make the effect perceivable
把原本 invisible 的 social effect 变得可见、可听、可感。因为人们低估自己的话对别人和对自己的影响。 (Cornell eCommons)

(c) Return value to the giver
让赞美者立即得到一种不是功利性的、但可感知的积极反馈：被放大、被转译、被保存、被回应。这样才能支撑你“赞美对双方都有利”的核心命题。 (PubMed)

三、Tentative design ideas

What kinds of interaction, form, or ritual can reflect the core concept and make the return of praise perceptible to the giver?

Design focus

Based on the theoretical framework, the design should focus primarily on:

- **social connection**
- **social value**

At the same time, the installation should preserve **autonomy**, so that praise feels voluntary, safe, and self-endorsed rather than forced.

Design Element 1: Preserving autonomy / voluntariness

Conceptual value:

> The installation should not force praise; it should create conditions in which praise feels safe, voluntary, and self-authored.

For example, instead of *Say a compliment now*，It could ask:

- *Think of someone whose presence made your day easier.*
- *Is there something you have noticed but never said?*
- *Would you like to leave a sincere appreciation?*

Possible formal characteristics:

- freedom to choose whom to address, what to say, and how to say it
- different modes of expression, such as speaking, writing, touching, or lingering

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Design Element 2: praise reduces social distance/making social connections visible

1. Two people (or one person with an installation）, or two ends of the installation, begin in a separate state. When one person expresses a sincere compliment, something in between them begins to connect: light, mist, lines, projection, or sound.

Possible forms:

- a line of light gradually connecting two points
- floating shapes moving closer and merging
- two projected circles slowly overlapping
- two separate sound fields blending into harmony
- guided movement in space （closer to each）

2. At first, each participant can only see part of the whole. When someone expresses praise, another part is unlocked, or a bridge appears between them. The full image or shared scene only becomes visible through expression.
3. Each compliment generates a visible “thread” between speaker and receiver. As more praise is expressed, these threads accumulate into a visible relational network.

Design Element 3: Making social value visible

1. A compliment does not disappear immediately after being expressed. Instead, it leaves a trace in space: something lights up, grows, accumulates, or remains.

- The speaker feels that their words carry weight
- the compliment affects a shared environment
- positive expression becomes something consequential rather than fleeting

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2. Compliments are anonymously preserved as part of a growing collective archive. The participant sees that their contribution is not just a personal moment, but part of a larger public space of mutual recognition.

- it transforms “my one sentence” into a social contribution

- it frames appreciation as a shared resource
- it reinforces the giver’ s sense of participating in something meaningful

Overall structure / procedure

Step 1: Notice （guidance）

The installation first slows the participant down and invites them to think of:

- a specific person
- a specific moment
- a specific quality or action worth appreciating

Step 2: Choose（autonomy）

The participant decides:

- whether to express praise
- whom to address
- how to express it

Step 3: Express （execute prosocial action）

The participant expresses a sincere compliment through some chosen mode.

Step 4: Receive the Return（social value）

The system transforms the compliment into a visible or tangible response:

- light
- sound
- spatial change
- trace
- physical artifact

Step 5: See the Link（social connection）

The participant becomes aware that something has changed not only outwardly, but inwardly and relationally.

For example:

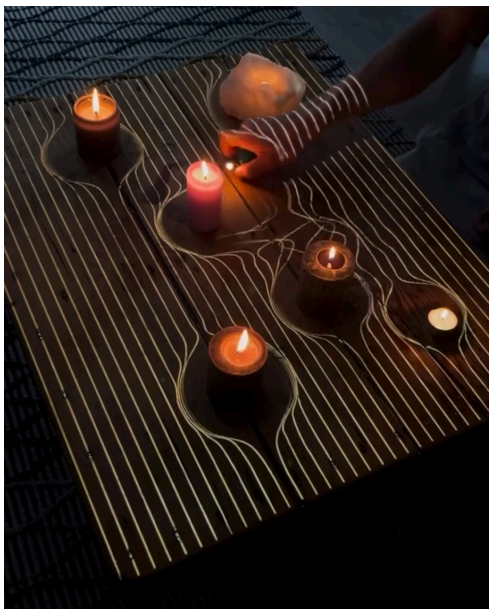
- They see a connection appear
- they see distance shrink

The theoretical framework and corresponding process have been determined, but the specific implementation form is still under consideration (choices can be made from the above examples, but should consider whether it is innovative and whether it can be realized).

四、Technical Report

A - 相照 (Mutual Illumination) **

- **释义：**出自“肝胆相照”或“交相辉映”。我们没有直接对话，甚至互不相识，但我们的烛火在桌面上却彼此相连、互相照亮。





Core Design Concept

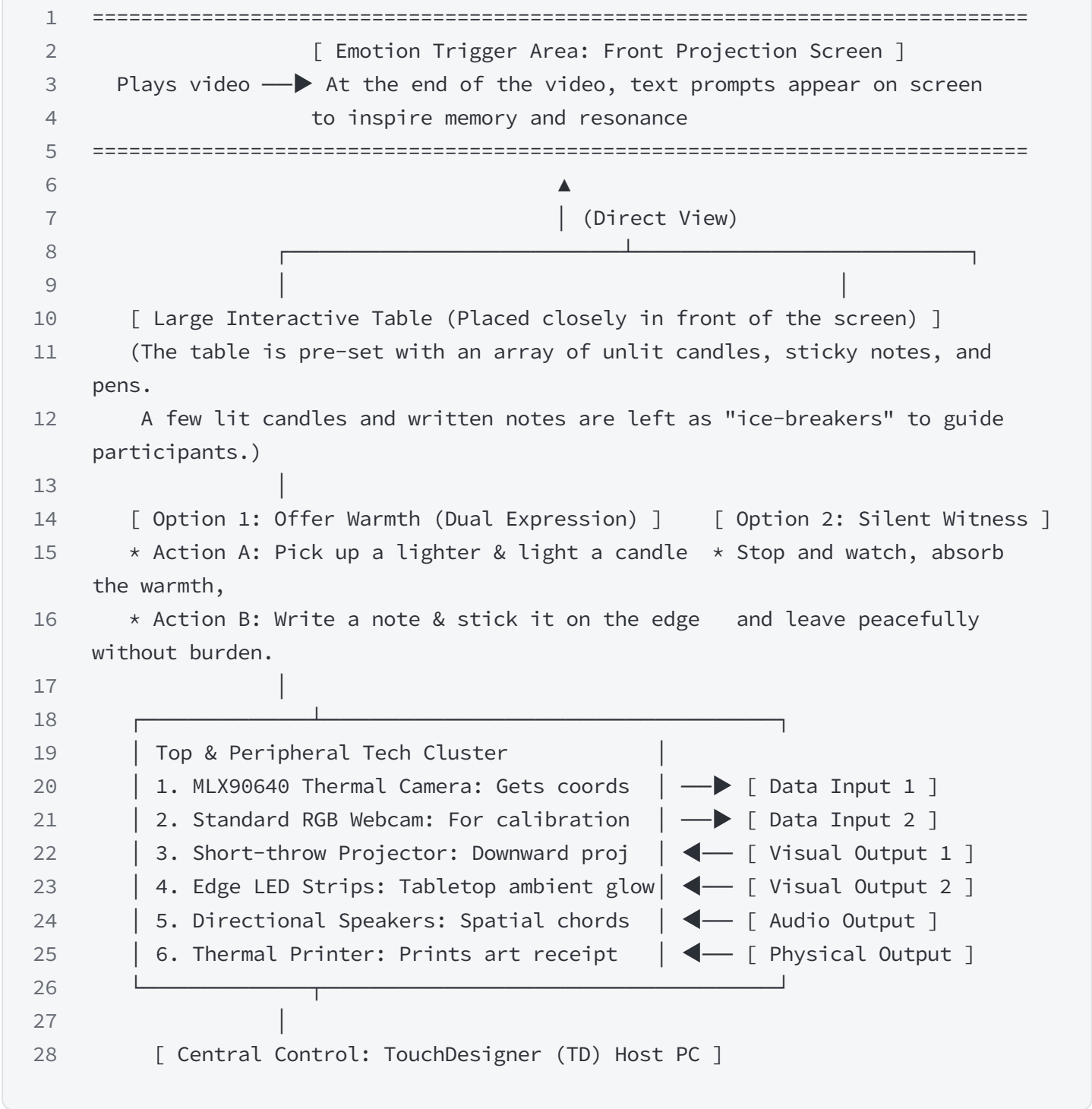
This installation transforms "praise and gratitude" from a potential awkward "verbal expression" into a poetic "transfer of heat" and "written record." A large interactive table is placed directly in front of a projection screen. Participants stand at the table to watch real, heartwarming moments collected from the community. When they feel empathy or resonance,

they can directly light a pre-placed candle on the table or write a specific compliment on a sticky note.

The system accurately captures the new physical heat source generated by the lighting action. Through a top-down projector, it generates visual dynamic light effects (ripples and connecting lines) around the real flame, accompanied by spatial sound effects and the glowing of ambient LED strips. Finally, the system captures the "network of light connections" belonging to the participant and prints a unique, generative art receipt for them to take away, completing a profound positive psychological feedback loop.

Spatial & Interaction Schematic

代码块



Detailed User Flow

Step 1: Immerse & Guide

Participants stand at the table and watch video clips on the screen. After the video ends, text prompts slowly appear: *"Please think of a specific person or event in your heart. Is there something you want to offer your kind praise to? If you can't think of anything right now, you can also praise the creator of this installation (e.g., 'Thank you for this exhibition,' 'Great job today')."*

- **Ice-breaker Design (Seed Content):** The interactive table is pre-set with a few lit candles and some written sticky notes on the table edge. This trace of "someone has already done this" provides a strong sense of safety and lowers the threshold for action.

Step 2: Express (Zero-Distance Interaction)

- **Minimalist Flow:** Unlit candles, sticky notes, and pens are right at the participants' fingertips. No walking is required; they just need to complete the core action—**Lighting** or **Writing**. Both can be done independently or simultaneously.

Step 3: Feedback (Light and Sound Effects)

When the thermal camera detects a newly added candle heat source on the table, TouchDesigner instantly triggers multi-sensory feedback:

- **Audio - Spatial Single Tone:** The system immediately plays a crisp, harmonious single tone (like a harp or wind chime).
- **Visual - Local Ripples & Connections:** The projector casts warm-colored water ripples around the new flame. At the same time, the projector slowly draws glowing connecting lines between the new candle and previously lit ones.
- **Visual - Ambient Elevation:** As the connecting lines form, the LED strips along the edge of the table emit a warm, breathing glow.

Step 4: The Takeaway

- **Generative Art Receipt:** After the system completes the connection animation, a thermal printer next to the table automatically prints a receipt.
- The receipt's pattern is a unique geometric connection map generated by the system based on the temperature and coordinates of the participant's lit candle. This proves that the light they gave left a mark, completing the psychological feedback loop in Self-Determination Theory (SDT).

Technical Implementation & Flowchart

The system uses TouchDesigner (TD) as the absolute core, with Arduino handling underlying physical data collection.

Dual Camera Configuration

1. **"Camera" for Coordinate Acquisition (MLX90640):** This is inherently a 32x24 pixel infrared thermal camera. TD receives a "black-and-white temperature map" from it, where only the high-temperature candle flames appear white (used to extract coordinates).
2. **"Camera" for Physical Calibration (RGB Webcam):** Because the digital light effects projected must perfectly align with the real candle flames, we install a standard USB webcam alongside the thermal camera at the top. It is used during the debugging phase to accurately align the virtual coordinate system in TD with the physical position of the real table (Mapping/Calibration).

Core Technical Details Explanation:

1. **Arduino Serial Communication:** TD has a built-in `Serial DAT` node that can seamlessly receive data sent by Arduino via `Serial.print()`. This wired connection avoids the potential lag and disconnections of wireless networks, making it the standard practice for commercial-grade interactive installations.
2. **Recognition of "New Heat Sources" (Baseline Subtraction):** After receiving the temperature frame, TD performs a background subtraction logic (current frame minus previous frame). The system only registers a "valid interaction" and triggers the sound and connection when the temperature at a specific location spikes within a short time (i.e., when a candle is just lit).
3. **Integration of Light Strips and Printing:** The edge light strips can be controlled by a second Arduino or an ESP32 with built-in Wi-Fi (flashed with WLED firmware). TD directly outputs DMX/Art-Net signals to synchronize the lighting effects with the projection. The printer achieves silent printing by invoking system commands through an internal Python script within TD.

[最难做的点 / 真实难点]

1. **热成像识别本身就有调试门槛**

MLX90640 不是“插上就能完美识别”。

你需要处理：

- 分辨率低（32x24）
- 热源容易互相干扰
- 桌面环境温度变化
- 火焰大小变化
- 阈值怎么设才不会误判

对于非计算机背景的人，这部分会比较吃力。
但它至少是**相对可查资料、可逐步调出来**的那种技术难题，不是抽象科研问题。

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2. **投影和真实桌面的对准（mapping/calibration）**

视觉效果要“刚好围着真实蜡烛出现”，这就要求投影坐标和热源坐标对齐。

这不是纯美术问题，而是实时映射问题。

难点包括：

- 摄像头视角和投影视角不同
- 桌面尺寸与投影尺寸要做校准

- 一旦设备移动，映射可能全乱

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3. **小票生成不是最难，但很容易变成杂活地狱**

打印看起来很浪漫，但实际上要处理：

- 截图时机
- 图形转黑白
- 热敏打印机驱动
- 静默打印
- 纸张卡顿

它不是科研难题，但很耗时间。

TouchDesigner Visual Requirement Brief

中文版｜TouchDesigner 视觉效果需求简述

1. 整体视觉方向

整体画面希望是干净、柔和、有发光感和空气感的。它不应该像 UI 界面，也不要像游戏技能特效。画面可以是较暗背景上的发光节点系统，带有轻微呼吸、缓慢流动和温暖氛围。整体感觉应该是安静、克制，但有生命力。

2. Node 光源点

每个光源点不要只是一个硬圆点，而应该像一个柔和的发光节点。它可以由一个较亮、较清晰的内核和一个更大、更模糊、透明度较低的外层光晕组成。节点可以有非常轻微的呼吸效果，比如亮度或大小缓慢变化，让它看起来像一个有温度的小光源。

3. Line 连接线

连接线不要是生硬的白色直线。它可以由一条较细的主线和一层更柔和的模糊光晕组成。线条出现时不要突然弹出，而是通过渐显或从一端慢慢生长出来。线出现后可以有轻微的亮度波动或缓慢流动感，但不要像高速电流。

4. Particle 粒子效果

粒子效果需要非常克制，只作为氛围辅助。可以是少量缓慢漂浮的小光尘、节点周围轻微扩散的粒子，或者沿连接线缓慢移动的小粒子。不要做爆炸、烟花、大量喷射、高饱和颜色或游戏魔法特效。

5. Field / Background 整体场域

背景不要是完全静止的黑底。随着光源数量增加，场域可以非常轻微地变亮。点与点之间或中心区域可以出现淡淡的雾气感 / 模糊光晕，像空气被轻微照亮。每次新连接出现时，整个场域可以轻轻 pulse 一下，但幅度要小，不能像闪屏。

6. 三种状态

当只有一个光源时，画面只有一个带光晕和呼吸感的 node，没有连接线，整体安静偏暗。

当有两个光源时，两个 node 之间逐渐出现一条连接线，节点光晕和背景亮度可以略微增强，并加入少量粒子。

当第三个光源出现时，它只连接最近的一个已有节点，不要立刻形成三条线。新的连接也要渐显或慢慢生长，同时整体 glow、haze 和粒子可以稍微增强，但仍然保持克制。

7. 过渡与优先级

所有状态切换都应该平滑，不要硬切。node 亮度、线条出现、背景变化、粒子数量和 haze 都应该缓慢过渡。实现上优先保证：node 视觉、两个点之间的连接线、第三个假光源与最近节点连接。其次再加入轻微场域提亮、简单粒子和淡淡 haze。

English Version｜TouchDesigner Visual Requirement Brief

1. Overall Visual Direction

The overall visual style should feel clean, soft, glowing, and atmospheric. It should not look like a UI interface or a game skill effect. The scene can be understood as a glowing node system on a dark background, with subtle breathing, slow movement, and a warm atmosphere. The overall feeling should be quiet and restrained, but still alive.

2. Node

Each source should appear as a soft glowing node rather than a hard circular dot. It can be built with a brighter and clearer inner core, plus a larger, more blurred outer halo with lower opacity. Each node can also have a very subtle breathing effect, such as a slow change in brightness or size, so that it feels like a small source of warmth.

3. Line

The connection line should not look like a hard white straight line. It can include a thin main line and a softer blurred glow layer underneath. The line should not pop up suddenly; it should gradually fade in or grow from one end to the other. After appearing, it may have a slight brightness fluctuation or a very slow flowing feeling, but it should not look like fast electric current.

4. Particle

The particle effect should be very restrained and only support the atmosphere. It can be a small number of slowly floating light particles, slight particle diffusion around the nodes, or tiny particles moving slowly along the connection line. It should avoid explosions, fireworks, heavy emission, highly saturated colors, or any game-like magic effect.

5. Field / Background

The background should not be a completely static black field. As the number of sources increases, the field can become very slightly brighter. A faint haze or blurred glow can appear between nodes or around the central shared area, as if the air is being softly illuminated. When a new connection appears, the whole field can have a gentle pulse, but the effect should remain subtle and should not feel like a flash.

Change the theme color(bg,node,line,effect)

6. Three Visual States

With one source, the scene should show only one glowing node with a soft halo and breathing motion. There is no connection line, and the overall field remains quiet and dark.

With two sources, a connection line gradually appears between the two nodes. The node glow and background brightness can become slightly stronger, and a small amount of particles can be added.

When the third source appears, it should only connect to the nearest existing node, instead of instantly forming three lines. The new connection should also fade in or grow gradually. The overall glow, haze, and particle effect can become slightly richer, but still restrained.设置一个可改参数，parametric setting。最好把所有东西都设置成variables。

7. Transition and Priority

All transitions between states should be smooth rather than abrupt. Node brightness, line appearance, background change, particle amount, and haze should all transition gradually. For implementation, the first priority is to make the node visual work well, make the two-source connection line clear, and integrate the third fake source so it connects to the nearest node. After that, subtle field brightening, simple particles, and faint haze can be added.

Purchasing / Inventory List

2. Complete Procurement List经过了chat和deepseek基于中国市场的询问

Recommended main chain: ESP32-S3 approach (total budget ¥400–700)

Category	Item	Recommended model / spec	Qty	Unit price (¥)	Subtotal (¥)	Notes
Sensing layer	Thermal sensor	MLX90640 (Waveshare brand)	1	250–350	250–350	Stable q
	Microcontroller	ESP32-S3 dev board	1	25–40	25–40	Choose comput
	Camera	Logitech C920 / compatible 1080p webcam	1	280–480 / 80–120	80–480	Big-bran
	Camera (depth alternative)	Intel RealSense D435 (bulk)	1	1,750–1,950	1,750–1,950	If budge
Output layer	LED strip	WS2812B 5V 60LED/m	3m	3–17/m	9–50	Cut-to-l
	LED controller	(ESP32-S3 doubles as controller, no extra purchase)	0	0	0	Flash W
	LED power supply	5V 3A adapter	1	10–15	10–15	
	Receipt printer	Xprinter XP-58IIH (USB)	1	80–150	80–150	Add last
	Thermal paper	57mm thermal paper rolls	several	1–2/roll	5–10	
Physical materials	Candles	Uniform spec tea lights (white, unscented recommended)	30–50 pcs	0.5–1	15–50	Same br consiste
	Long lighter	Long-handle lighter	2	5–10	10–20	Safer

		lighter				
	Heat-resistant tray	Aluminum/steel large tray	1	30–80	30–80	Fire safe
	Black matte surface	Black matte PVC sheet / cardstock	1 roll	20–50	20–50	Reduces
	Sticky notes + pens	Materials for writing praise	several	10–20	10–20	
	Mounts / stands	Camera tripod / C-clamp	2–3	50–100	100–300	For mou
Software	TouchDesigner	Non-Commercial license	1	0	0	Free for

> Total without D435: approx. ¥519–1,395 (excluding borrowed projector), very budget-friendly.
> Total with D435: approx. ¥2,269–3,345, close to SWRG ceiling.

Data flow diagram

代码块

```
[Candle burning]
      | Infrared radiation
      ▼
[MLX90640 thermal sensor]
      | I2C
      ▼
[ESP32-S3 dev board]
      | (1) MLX90640 library reads 32×24 temperature matrix
      | (2) Threshold filter → extract new heat source coordinates (x_idx, y_idx)
      | (3) Serial.print() sends via USB Serial
      |
      ├──────────────────────────────────┐
      ▼                                  ▼
[TouchDesigner PC]              [WS2812B LED strip]
Serial DAT receives             (ESP32-S3 flashed with WLED,
→ CHOP parses x, y, temp          controlled via Wi-Fi or serial)
→ TOP/SOP generate visuals
      |
      └──────────┬──────────┐
                 ▼          ▼
            [Projector]   [Speaker]
```

```
                (HDMI)       (3.5mm/USB)

  (Final stage add:)
  [TouchDesigner]
  Python Script → screenshot → [USB thermal printer]
```

Technical Feasibility Checklist

Function module	Technical approach	Feasibility	Primary risk
Candle detection	MLX90640	✅ High	Multiple candle interfering; amb change
Heat coordinate extraction	Threshold filtering + baseline subtraction	✅ High (gradually tunable)	Single source m sources need fir
Data into computer	ESP32 USB Serial (or M5Stack Thermal2 direct USB)	✅ Very High	Serial communi method
TD visual generation	TouchDesigner	✅ High	TD is the most s project
Projection alignment (webcam approach)	Webcam + OpenCV manual calibration	⚠️ Medium	Recalibration ne risk of visual err
Projection alignment (D435 depth approach)	D435 depth camera SDK to get 3D coordinates	✅ High (more stable)	Need to learn D4
Borrowing projector	Borrow from university	✅ High	Confirm availab
LED strip	WS2812B + WLED	✅ Very High	Plug and play
Receipt printing	USB POS printer + Python	✅ Medium	Driver and scree debugging

导师建议用来对齐的camera可以先不买，到时候试试手动眼睛看着对齐，或者拿手机摄像头先充当一下，再看

蜡烛也许可以买有味道的。

0824 review

Mutual Illumination / 相照 — Capstone Fall Restart

	中文	English
初衷 / Motivation	我最开始关注的是：**人们经常注意到别人值得被赞美的地方，但不一定真的表达出来。** 原因可能是尴尬、担心显得不真诚、不知道怎么表达等。因此我想探索，交互设计是否可以降低 praise 的表达门槛，同时让这个行为本身对给予者也产生可感知的意义。	I started from the observation that **people things they appreciate about others, but necessarily express them.** They may feel a about sounding insincere, or simply not kn I wanted to explore whether interaction des this barrier while making the act of giving p meaningful to the giver.
核心研究问题 / Research Question	**如何通过交互装置，让给予 praise 所产生的 positive return 被 giver 自己感知到？** 我关注的不只是 receiver 得到了什么，而是 giver 在给予 praise 后获得的 **social connection、agency 和 social value**。	**How can an interactive installation make return of giving praise perceptible to the** focusing only on the receiver, I am intereste praise can return a sense of **social connect social value** to the giver.
Warmth Metaphor	核心 metaphor 是 **warmth**。Praise 经常被描述成一种“温暖”的行为，所以我把抽象的 social warmth 转化成真实的 physical warmth：**点燃蜡烛 → 产生热量 → 系统检测 → 转化成光和连接。**	The central metaphor is **warmth**. Praise is as something warm, so I translate abstract into physical warmth: **lighting a candle → → detecting it → transforming it into ligh connection.**
最终体验 / Experience	每一次 praise / 点燃蜡烛都会生成一个 glowing node；新的 node 出现后产生 ripple，并逐渐与其他 node 建立连接。随着参与者增加，独立的行为逐渐形成一个共同的光网络。**我真正想可视化的是 praise 给 giver 带来的 return，以及个体之间原本不可见的 social connection / social value。**	Each act of praise / candle lighting creates new node produces a ripple and gradually existing nodes. As more people participate, actions form a shared network of light. **Wha to visualize is the return of giving praise otherwise invisible social connection and between people.**

目前已经做到哪里 / What I Have Done

1. **Research & Concept**
   - Questionnaire-based preliminary research
   - Literature review + theoretical framework
   - Interaction concept （warmth metaphor）
   - Technical workflow planning
   - scenario expectation / visual direction
2. **Prototype（2）**
   - **TD Prototype 1 — Detection:** 用 camera + phone flashlight 模拟 heat source，通过 brightness threshold + Blob Track 得到坐标。
   - **TD Prototype 2 — Visual:** 做了 glowing nodes、lines、particles、field effect，以及简单的 multi-point simulation。
   - **目前的问题：这两个 prototype 还是独立的，还没有真正 integration。**

Summer holiday Update

- **SWRG funding:** I submitted the funding application, but **I haven't received any follow-up or response after submission.**
- **IRB: Approved.** So formal user testing can move forward later according to the approved protocol.
- Funding 申请里原本计划的核心硬件是 **MLX90640 thermal sensor + ESP32-S3**；LED、thermal printer 等属于可以后加的 secondary components。

Timeline / 我希望这学期怎么推进

> **My goal is to have the physical installation built and mostly debugged by the end of October.**
>
> I don't want to use the official capstone deadline as my actual production deadline, because I need this project for my graduate school applications. Ideally, after October, I want to shift from building the system to **user testing, refinement and portfolio preparation.**

我现在的状态 / Where I Am Now

> **I know what the final direction is, but I'm a little unsure about the most efficient way to move from these separate prototypes into a working physical installation.**

> Right now, I think the next step may be to connect the detection prototype with the visual prototype, and also start purchasing and testing the real thermal sensor and ESP32.
>
> I'm also wondering when I should start testing multiple input points. My current free TouchDesigner setup only allows me to work with up to two detected points at the same time, so I'm not sure whether I should solve that limitation now, or first focus on getting a simple two-point physical prototype working（**minimum working pipeline**）.

我个人会建议你的推进顺序不要是：

买齐所有东西 → multiple points → LED → printer → projection → visual……

这样又会乱。

而是非常暴力地先做一条 **minimum working pipeline**：

1 Real candle → Thermal sensor → ESP32 → TouchDesigner → Projector → projected glowing node appears around the physical candle

↓

2 candles → connection

↓

multiple points / richer visuals

↓

secondary outputs（LED / sound / printer）

What's the next step?

1. Move from 2-3 nodes（the line connection、effects）， whether it is stable、perceptible
2. copy（尝试用Github repo留痕，随时可回退）
3. Visual is a huge process，use own language

视觉、材料意向：

流动、传递（线的流动、颜色的改变）

蜡烛上作文章：外部形状、燃烧的痕迹、燃烧过后留下的东西

意义：传递、延续、给予温暖不等于消失或消耗





和镜子的搭配可以用蜡烛灯替代、珍珠蜡、冰花蜡、沙蜡（传递、融化/融合、凝聚：每一粒沙是一个个体，通过赞美让大家凝聚形成了一个connection的感觉）、还能留下痕迹（被提起来）